

SmRF — Smart Radon Fan

Category Topic: Can I trust SmRF??



Smart Radon Fan

SmRF includes fault-tolerant features

- Goal: SmRF-enabled radon mitigation is as reliable as a standard radon fan installation
- SmRF builds on top of standard radon mitigation systems
- SmRF improves radon mitigation visibility



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SmRF Fault-Tolerance Summary

Failure Type	SmRF Behavior
Airthings sensor failure	Fail active (fan forced ON)
Fan control-path failure	Fail visible (UNKNOWN state + alert)
Internet/cloud failure	Local autonomous operation continues
Home Wi-Fi/router failure	Local operation continues when possible
Pi IP address change	AP interface reveals new IP
Shelly IP address change	Automatic rediscovery and reconnection
OTA updates	Signed updates + rollback support
Power interruption	Automatic reboot and recovery



SmRF is designed so that failures degrade toward standard radon fan operation

The household loses power

- The radon fan, if active, stops as before
- When power is restored, SmRF will reboot and return to its pre-interrupted state
- The SmRF Back Office (=SmRF Viewer) reports the installation as offline



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The SmRF Raspberry Pi fails

- SmRF Viewer reports the installation as offline



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SmRF loses Internet connection

- SmRF Viewer reports the installation as offline
- Locally, the SmRF continues to run, and may continue normally
- When connectivity is restored, SmRF re-connects automatically



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Home WiFi network fails

- SmRF Viewer reports the installation as offline
- When connectivity is restored, SmRF re-connects automatically



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Home WiFi router changes device IPs

- New SmRF Raspberry Pi IP is available from the smartphone AP interface
- The SmRF Viewer is updated with the new Raspberry Pi IP
- The Shelly Smart Plug is auto-rediscovered by SmRF



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Airthings CRM can't be read (or produces questionable readings)

- Radon fan is turned ON as default
- SmRF UI shows failure
- SmRF Viewer is informed



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Shelly Smart Plug can't be accessed

- SmRF UI shows failure
- SmRF Viewer is informed



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Worst-case scenario

- SmRF Viewer reports the installation as offline
- The Smart Plug is removed, and the fan is plugged directly into the fan receptacle



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Fault-tolerant Testing

- Development testing of identified failure modes
- In-field Beta test results



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Let **SmRF**
protect your home



Know Your
System is Working



Automated
Radon Mitigation



Real-Time
Feedback & Control



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Thanks for taking a look at SmRF!